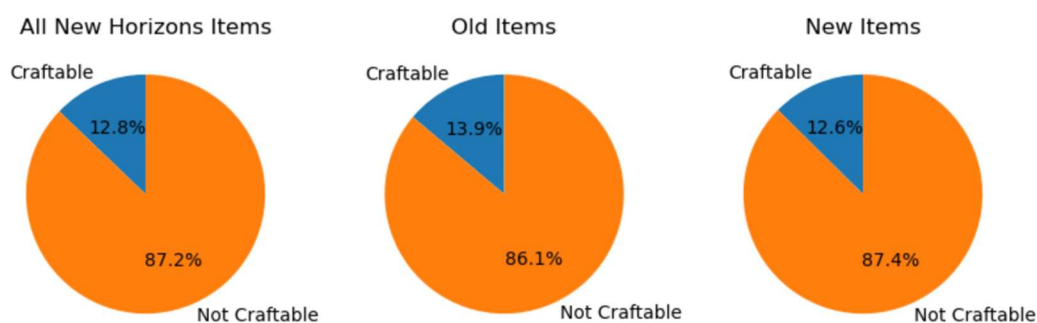


For my project I decided I wanted to look at the video games Animal Crossing New Leaf and Animal Crossing New Horizons and see if by analyzing the data relating to the games items if I could find any key differences in how the two games were designed. To give a quick summary Animal Crossing is a series of social simulation games in which you as the player have just moved into some town and are tasked with becoming friends with your neighbors, decorating your home and the rest of the town, and catching various creatures. Animal Crossing New Leaf (released in 2012) shook up that formula slightly by making you the mayor of your town thus letting you oversee more large-scale projects while Animal Crossing New Horizons (The most recent game in the series released in 2020) had your moving to a deserted island and turning into a proper town. My reason for choosing this as my topic is that I'm a fan of the Animal crossing series and there is a frankly ridiculous amount of information about this franchise and all the games in it that have been archived by fans of the franchise making getting data for this project pretty easy.

All the data used for this project was taken from the Animal Crossing New Horizons and Animal Crossing New leaf spread sheet projects which are a part of a series of projects that try to catalogue everything there is to possibly catalogue about these games. These projects are generally agreed to be incredibly accurate but there is a chance that some of the data is wrong as this data comes from fans of the games and not an official source and I did not double check the data with the games themselves. For my project I have used the spreadsheets cataloguing all of New Horizons and New Leaf's purchasable items and creatures. For cleaning the data, I had to do very little for the New Horizons spread sheets but for the New Leaf ones every instance of an empty cell contained "-" which I had to replace with NAN.

New Horizons added a new mechanic to the series in crafting which allowed you to combine resources together to make tools, clothes, or furniture instead of having to buy them with bells (the in game currency). Not all items in New Horizons though can be crafted so I wanted to see first what percentage can be and then see how many of the new items introduced to New Horizons can be crafted compared to how many of the old items from New Leaf can be crafted. To do this I created the following three pie charts:



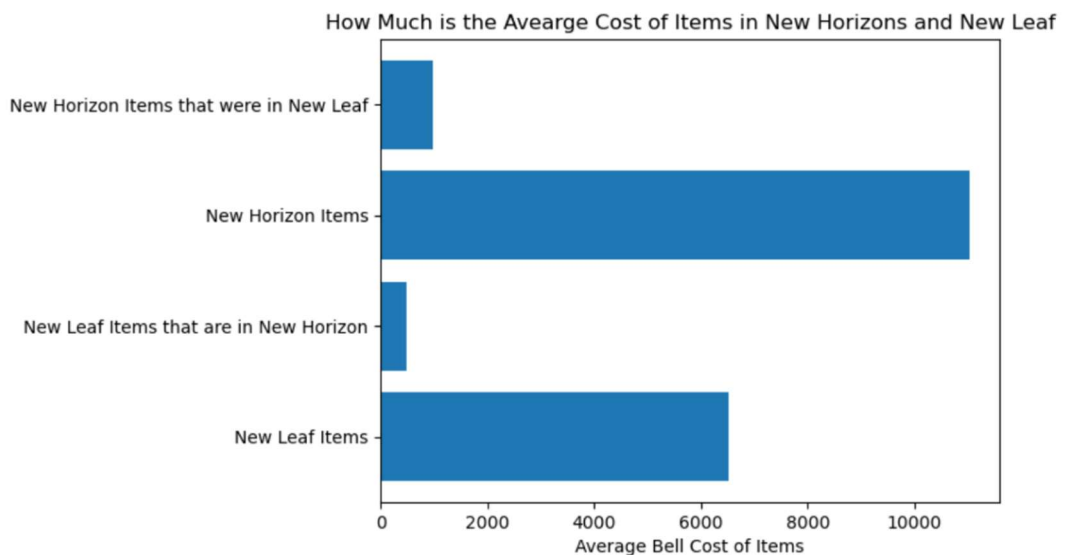
For the first one I just looked at all the items from New Horizons and saw if they were craftable or not. For the second and third chart though I first had to either remove all the items also in New Leaf or that weren't in New Leaf. From these charts we can see how all three pie charts are very similar with around 13% of the items in each of the categories being craftable. This surprised me as I expected there to be first, a much higher percentage of craftable items in the game and second, for the new items to have the highest percentage craftable out of the three as I had assumed that the team working on New Horizons would have wanted to push players to spend more time crafting as that was a new gameplay addition to the series ,and that most of the new Items created for New Horizons would be craftable as a result. But it seems I was completely wrong in that regard with instead the old items having the highest percentage of craftable Items. This points to the development team of New Horizons not wanting to shake up the pre-established gameplay formula of the series as I much as I initially expected.

The next thing I wanted to see was how much the average amount of bells items cost in New Horizons compared to new leaf which I found by taking all the items from the games and how much those items cost to buy from the games various shops and dividing it by the number of items in the game. I did also have to exclude some of the items that didn't have a cost because they are obtained some other way and the items that have random prices. From

that process I got this:

From This bar graph we see how not only us the average price of items in New Horizons nearly twice as high as the

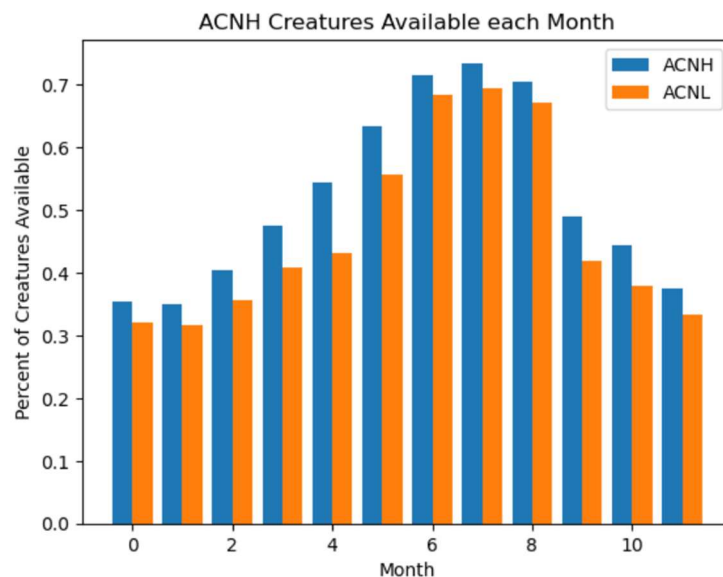
price in New Leaf but that also if we just compare the items that were in both games the price is still nearly double. This does make sense though as New Horizon added a few new ways to earn bells that weren't in New Leaf such as exploring random islands and selling crafted items. So, it is likely that to compensate for the fact that players would be making more bells the average price of items was increased to counteract that. This would in theory, I think keep the pacing at which you get new items in New Horizon similar to New



Leaf showing how the development team of New Horizons was trying to keep the games similar in that regard.

In New Horizons and New Leaf you can fish, catch bugs, and dive for sea creatures to sell or display. The creatures that are available for you to capture though are based on the real-world date with only certain creatures available each month. So, I wanted to see how many creatures are available each month for both games to see if they changed at all. To do this for the New Horizons creatures I was able to use the spreadsheet as it had a column for each month that would have a “yes” in it for if the rows creature was available then, so it was simple enough to tally up the creatures for each month. Though I should also specify that for New Horizons the creatures available each month are different if you’re in the northern or southern hemisphere, because this was not the case in New Leaf which always defaults to being in the northern hemisphere I decided to compare the New Leaf creature data to just the northern hemisphere data for New Horizons. For the new leaf data though it was far more difficult to use the spreadsheet directly with python to tally the creatures for each month because the New Leaf spreadsheet saves the availability as a single column with text in it describing the month availability of the creatures. Because of that I opted to just tally the New Leaf creatures manually while using the data from the spreadsheet. Doing that I got the following:

From this bar graph we can see that the trend of when creatures are available is roughly the same for both New Horizons and New Leaf. With New Horizons having a higher percentage of its creatures available more often. The availability also trend higher in the summer probably because more animals tend to be active when its warmer out so it would make sense for more creatures to be available and because children who are the primary demographic of the abnimal crossing series would likely have more free time to play and catch all the animals in the summer.



From what I’ve found in doing this research it seems that the people working on New Horizons were very likely trying to keep the overall pace of the game and its progression similar to that of New Leaf. The percentage of creatures available each month being roughly the same shows how the development team wanted/expected people to play New Horizons a similar amount each month as New Leaf. The changes to item cost were most

likely done to counter act the new ways they added to earn money in New Horizons and the surprisingly low number of craftable items which was one of the big new things in New Horizons shows that despite clearly wanting to innovate in some aspects with the new gameplay mechanics they introduced (crafting, island exploring, a new setting) they still clearly wanted to keep the game feeling similar to their previous work New Leaf. If I was to continue this research, I could take a look at the older Animal Crossing games and any future ones and compare information from them to what I've already found. Or even compare this information to other series in social sim genre to see how those games are designed differently.